

Santhosh G BE(EEE)
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OBJECTIVE

I am seeking a creative and challenging environment where I can share my innovative ideas, leverage my experience, and take on full responsibilities in my role.

CAREER PATH

5+ years of experience in Embedded Software Development, specializing in STB, dash cameras, RISC-V platform bring-up, bootloaders, upgraders, middleware, and WebKit browsers, with a focus on continuous learning and improvement.

PROFESSIONAL EXPERIENCE

Senior Software Engineer–Tessolve Semiconductor Pvt. Ltd, Bangalore (*Aug25–Present*)

Associate Software Engineer– Hero Electronix Pvt. Ltd, Bangalore (*Jan25–Aug25*)

Associate Software Engineer– SureSoft Systems Pvt. Ltd, Pondicherry (*July21–Jan25*).

TECHNICAL SKILLS

- Proficient in **C**, **C++**, and **shell scripting**. Debugging skills in **JS** and **HTML**.
- Experienced with **Makefile (GNU Make)**, **CMake**, and auto-build environments using **Python** and **Jenkins**.
- Experienced in **RTOS**, **Linux** and **socket programming**.
- Skilled in platform bring-up for **bootloaders**, **upgraders**, **HAL** and **Middleware** software porting for various vendors.
- Knowledgeable in chip-specific software **encryption and decryption** methods.
- Proficient in **HbbTV**, **DIAL**, **CURL**, **analytics**, **web remote**, **HTML5** and **MSE**.
- Experience with various microprocessors and microcontrollers/toolsets including **MediaTek**, **Montage**, **Ali**, **HiSilicon**, **National**, **SigmaStar** and **NextSilicon**.
- **Yocto build system**: recipe creation, rootfs customization, and full system integration.
- **RISC-V platform experience**: board bring-up, U-Boot porting, and driver development.
- **Linux programming expertise**: processes, threads, message queues, shared memory, FIFO, and semaphores.
- **Device experience**: IRRX, LED, FPD, NOR, NAND, eMMC and NVMe.
- **Familiar with protocols**: SSDP, TFTP, NFS, TCP, UDP, HTTP(S) and PCIe
- **Utilities**: Source Insight, SVN, GIT, Hex Workshop, and REST client/server setup.
- **Debugging tools**: GDB, Valgrind, Doxygen, and Visual Studio Code.
- **Fusing tools**: MStar ISP Tool, Ali WinGDB, and Montage SCOM Tool.
- **Bug tracking tools**: eService, Mantis, Redmine, and WIAS (SureSoft proprietary).

PROJECT EXPERIENCE

Tessolve Semiconductor, Bangalore, India(*Aug 2025 - Present*)

- Performed **full board bring-up** for next-generation RISC-V silicon.
- Ported and customized **U-Boot**, including driver-level modifications.
- Implemented **PCIe-based NVMe boot support** for high-speed storage.

- Integrated and customized **Yocto build system** with prebuilt rootfs deployment.
- Conducted platform validation using **CoreMark, CoreMark-Pro, and stress-ng**.
- Debugged early boot issues across **firmware, bootloader, and kernel layers**.

Hero Electronix Pvt. Ltd, Bangalore, India (*Jan 2025 – Aug 2025*)

Project: Dash Camera

- Set up **bootloader and middleware build environment**.
- Implemented features such as **towing detection, idling detection, and OTA updates**.
- Debugged and fixed issues in **GPS and G-sensor modules**.
- Enabled **triple-camera recording system integration**.

Suresoft Systems Pvt. Ltd, Pondicherry, India (*Jul 2021 - Jan 2025*)

Project: WebKit Browser

- Ported and integrated multiple versions of **WebKit (process & non-process models)**.
- Implemented **HbbTV lifecycle and MSE playback support**.
- Developed multi-instance browser support for **IP channels and portal apps**.
- Integrated **DSMCC-based data handling**.
- Debugged rendering, playback, and lifecycle issues.

Project: Bootloader & Upgrader

- Integrated **IRRXX**, front panel, LED, keys, and boot logo into the bootloader.
- Implemented secure boot features using **CRC, RSA, MAC, and digital signatures**.
- Designed **TEE execution flow within REE loader**.
- Developed upgrade mechanisms: **USB, OTA, OND**.
- Implemented **aligned/unaligned read-write operations for eMMC**.
- Defined **MTD partitions and boot arguments**.

Certification

- Verimatrix Loader Certification
- Irdeto 5.1 Secure Bootloader Certification
- BBC Certification.